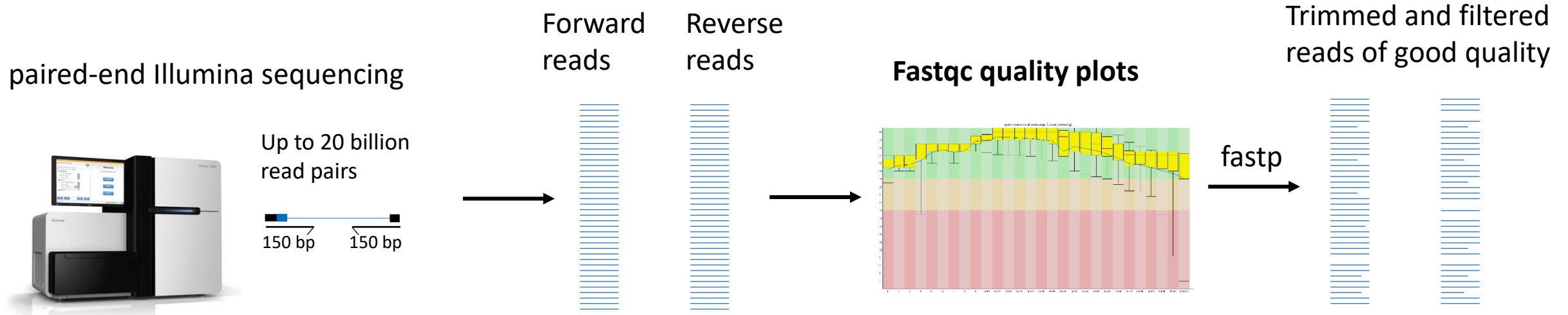


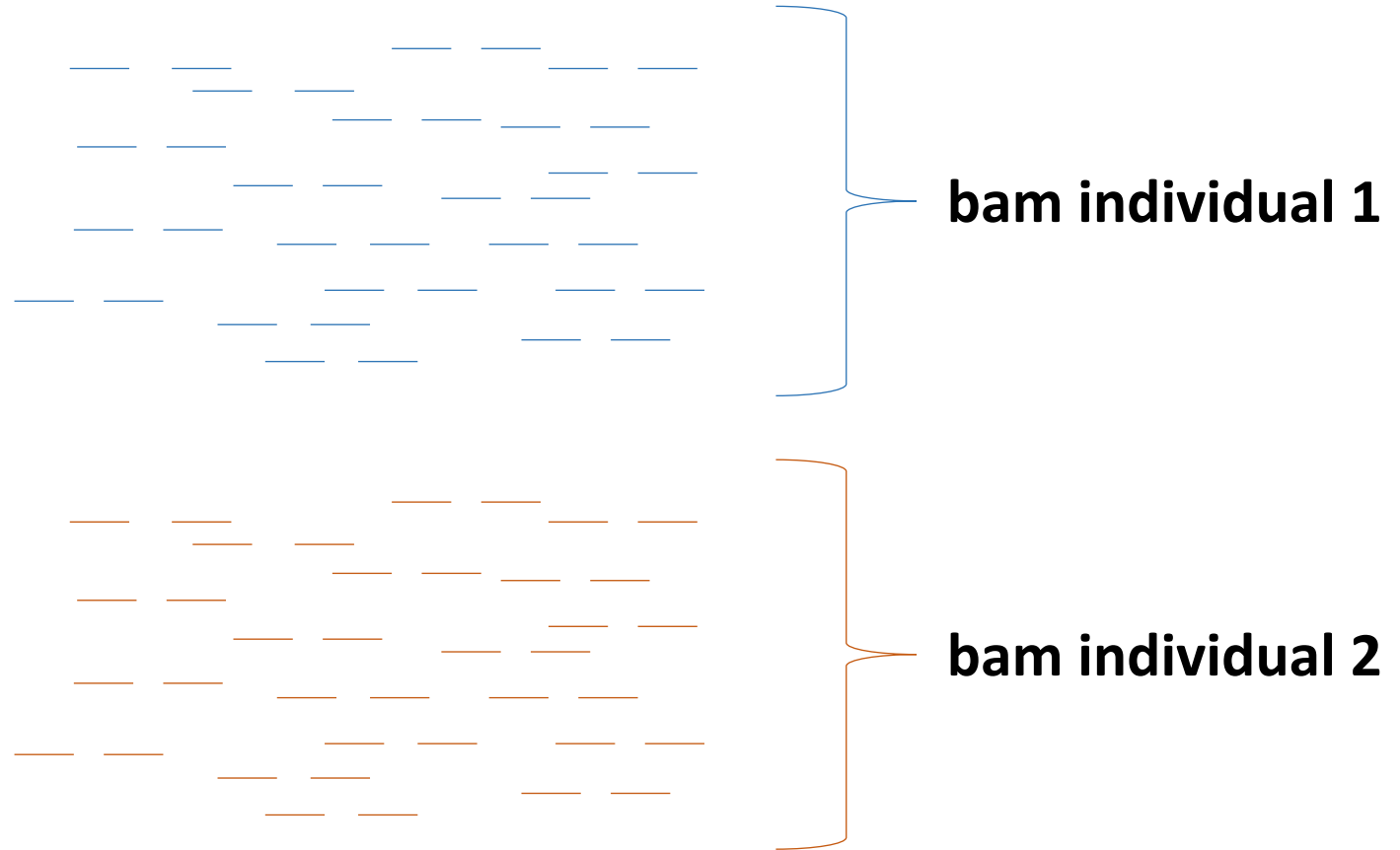
Mapping to a
reference genome

Yesterday: Quality check and trimming raw reads



Today: Alignment to the reference genome

reference



Aligning/Mapping short read data to a reference genome



bwa-mem2 (Burroughs-Wheeler Alignment with maximum exact matches)

Version 2 is 50% faster than bwa-mem

Very new tool worth considering: minibwa
(<https://arxiv.org/abs/2606.15357>) which is much faster, similar quality
and even better at aligning reads with indels

Regions that are repeated in the genome make it very difficult to map reads

Reference genome



Read to be aligned to the reference



Regions that are repeated in the genome make it very difficult to map reads

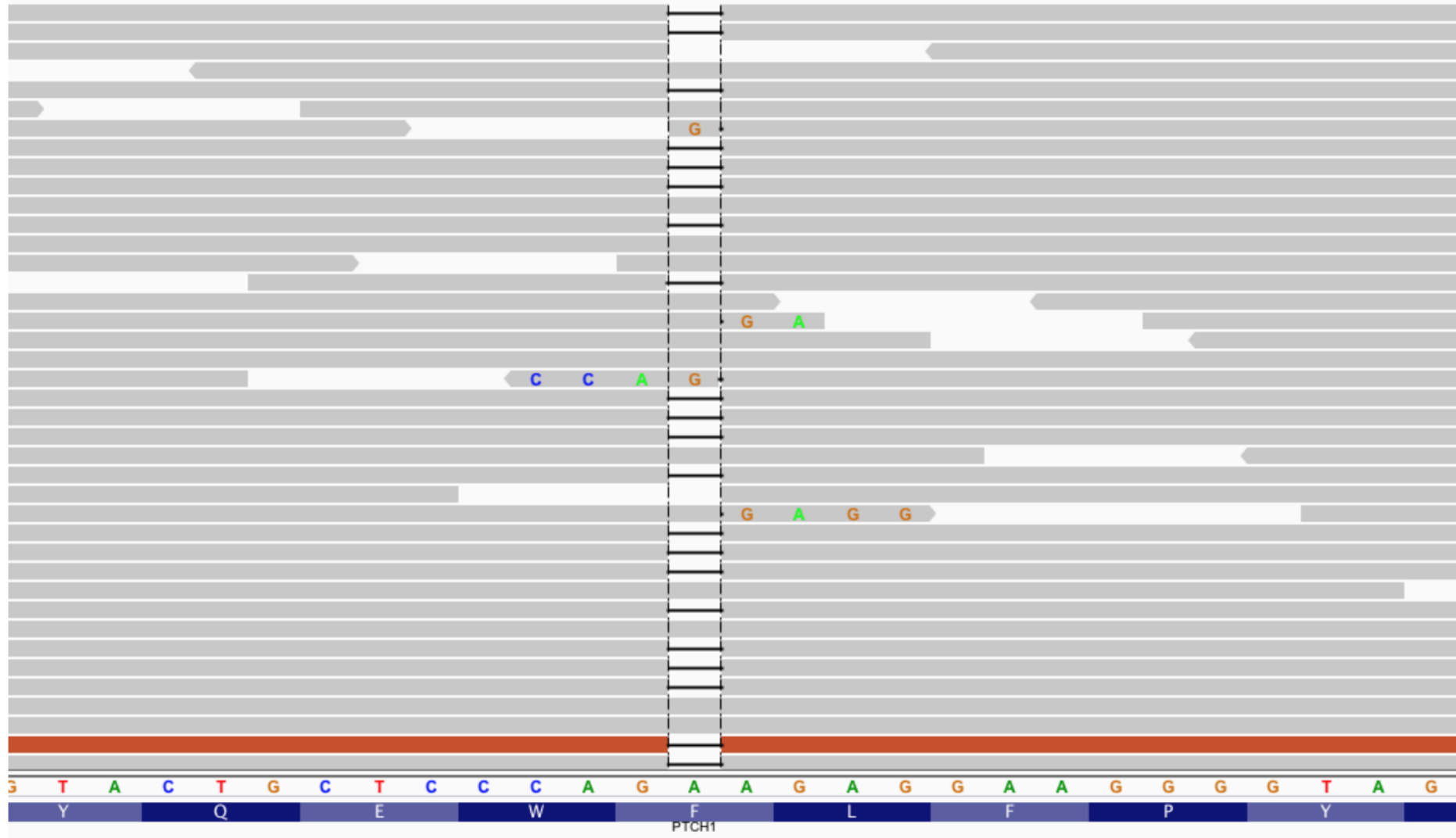
Reference genome



Read to be aligned to the reference

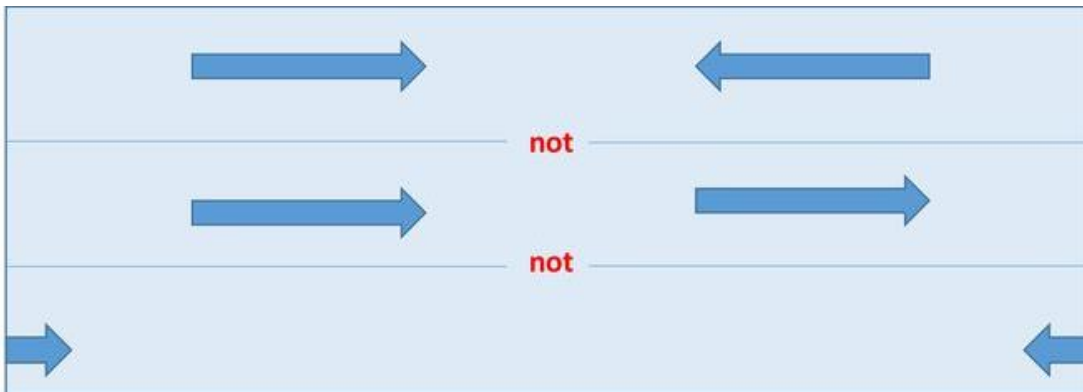


Indels: deletions and insertions



Aligning reads is complicated by:

- Imperfect match to the reference due to mutations or sequencing errors, or errors in the reference genome
- Multiple positions where the read could match (repeated regions)
- Low quality of the read
- With paired-end reads: only one read maps or the other read maps on a different chromosome



Alignment tools such as bwa-mem2 are able to handle all of these complications and give information on the mapping quality.

General principle

- The reference genome is indexed to make mapping faster
 - Reference: ATCATTCATCTG (would of course be much longer)
 - Index: for each k-mer figure out where they are in the genome
e.g. for k=3, k-mers are combinations of 3 nucleotides

Positions of 3-mers in our toy example:

ATC: 1,8 (ATCATTCATCTG)

TCA: 2 (ATTCATTCATCTG)

CAT: 3,7 (ATCATTCATCTG)

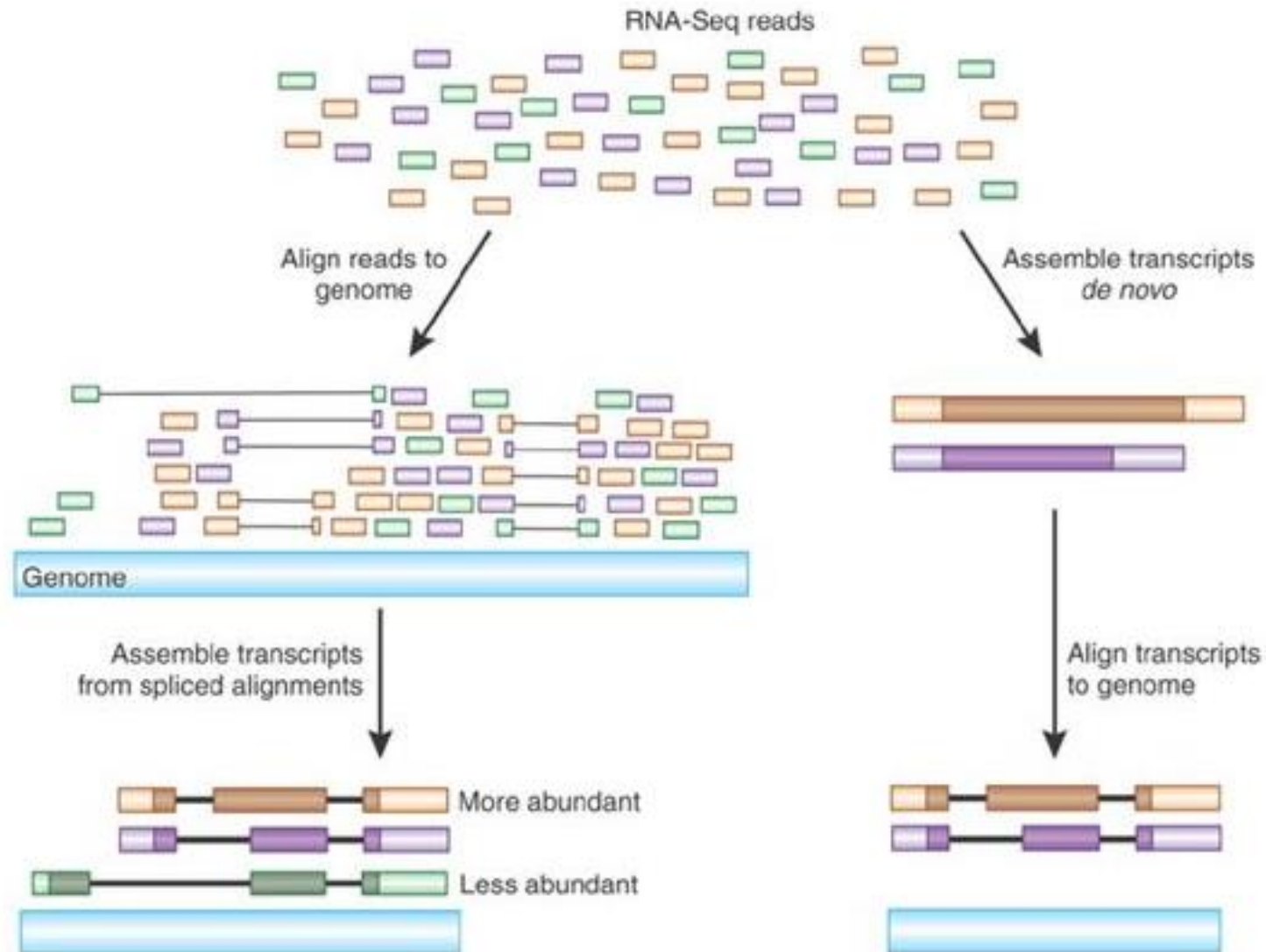
ATT: 4 (ATCATTCATCTG)

etc.

- The reads are mapped using the index:
 - Read example: CATTC
 - First k-mer (CAT) suggests the read could start at position 3 or 7
 - Second k-mer shows that it matches (ATT) at position 4

Aligning other type of data: RNA data

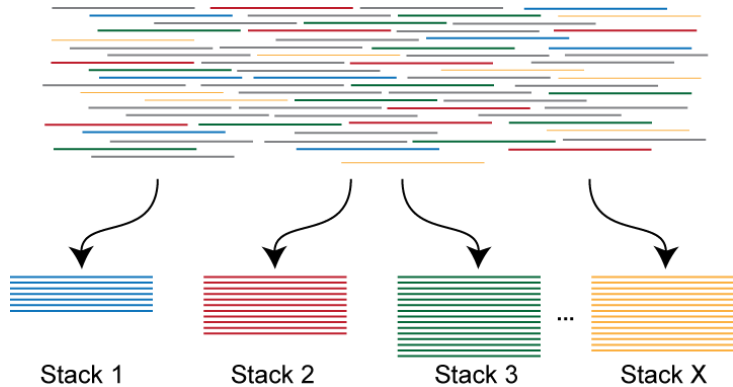
Tools:
e.g.
STAR
HISAT2



RAD data with or without a reference genome

Make your own reference with:

- **STACKS**: for datasets of <https://catchenlab.life.illinois.edu/stacks>



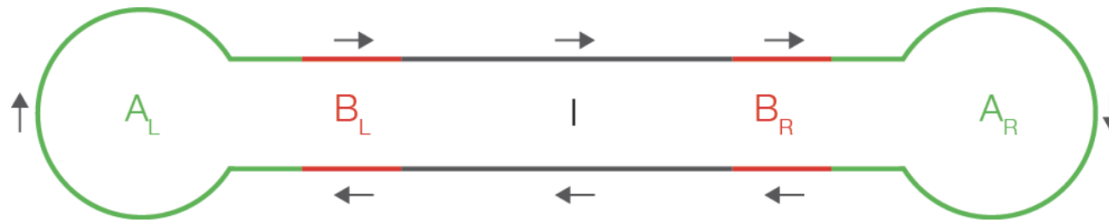
 **Stacks**

- **iPyRAD** (for RAD, ddRAD, GBS, 2bRAD): for datasets with different species <https://ipyrad.readthedocs.io/en/master/index.html>

Long read mapping

Karin Näsval

Mapping long reads PacBio



A schematic drawing of a SMRTbell

A =Adaptors, B=Barcode, I=insert (DNA of interest)

Automatic HiFi reads generation (Export Reads)

`hifi_reads.fastq.gz` - FASTQ file containing HiFi Reads
`hifi_reads.fasta.gz` - FASTA file containing HiFi Reads
`hifi_reads.bam` - BAM file containing HiFi Reads

HiFi reads QV > 20

<https://pacbiofileformats.readthedocs.io/en/13.0/>

PacBio

```
cat m84093_240426_124306_s1.ccs.bc2033.stats
```

```
-
```

```
A = 7108650862 (34.9%), C = 3058767823 (15.0%), G = 3061827434 (15.0%), T = 7127816130 (35.0%),  
CpG = 1003156910 (4.9%)
```

```
sum = 20357062249, n = 2218131, mean = 9177.57438537219, largest = 30301, smallest = 135
```

sum = total number of bases

n = number of sequences

mean, largest, smallest = length of reads

PacBio

<https://pacbiofileformats.readthedocs.io/en/13.0/BAM.html#hifi-reads>

m84093 240426 124306 s1/261297081/ccs 4 * 0 255 * * 0 0

GTTTGGTTATATTGTTCATAGACACCTACGGTACACGAATATAATAAACAAATAATATTAATTTGTTATACATATTTAAATATAAAATTAATATTGTAGCAACTATTAACATACTAGTTTAC
 TTCCTTCAACAGTTTAAATCAAGGATTCTTAACATTTTTTACCAAACGTAAGTATACCAAGATTGTTTTCTAATGATTTTGATGATATGGATGTGATTTGACTACATTAATGCTA
 CTGCTAGAACTTTACGATATCCCTCACATCACCTCGTCAAATGCTTTGTTTAAATTGTTAAAAAAATTCTTTGTATTTAAAGTGAGTAATCCTTAAGTAAACAGTGTTTCAAAGT
 TAAAAATTTATTTTATACGTGAAATGTCGTATTATCGATGGCATTATTAGTTTCCAGCATATTTATAAAAGCCTTTGTGACGCAATCACCGAGCGGTGAAAATATGATTTGAGACGCT
 TCCTGAAATTGAAAAAAGAAACCCCTCTATTGTTTTTAGATGATGTTTATTTTCTGCATTACTTTCTTAAATTAATATAATTTATACTATACAAATGTGACATTTCAATTGCGATACGA
 GTTTAAGAGGAGAACATCGATTACTGAAGTTATACAAACTTCTCCACAATAGATAACATTAACCTTTATATTTTAAACTTTAAGATAGTTCTTAAAAGTGTGAAAAGATGTAAAAAT
 GGTACATTTGAAGAAACCTTTATTTAAGAGAGTTTAAAGGTATCGGTATTTTCAAAGAGATTTTACATCTTTCTAATGCGAAATGTTAAAGATCTTAAAAATATTTTTAGCTTAT
 TTAACGAATAAGAAGAGAAAATATTTATTAATAATTCATATTTGATATTTATATTGATTTTATTTATTTCTTTTAAATAGGTTTAAATTAAGTTTCGATTTTCAAATCTCTTTTATTGT
 TAAATGTTTCATCTGGAGACATTTACAGTCCAATAGTTTCCAAAAGAAATTTTAAACGCTGTTTCGAAATATCTTTCTGTGATTATGAAGTAACTGTTAAAAATTCCTACTCAGTTAATG
 GTTTAAAATTTTATGTATTGCAAGAAAAGATAATAAAAGCTTGAAATAAAATCATTTTATCAGTATTTTGCATACCTATAAATATTTGAAGGTA

— — — —

[illegible]

ac:B:i,8,0,10,0 bl:Z:CGACTCGTATGAGTAI bt:Z:ATACTCATACGAGTCG bx:B:i,16,16 ec:f:7.5012 ma:i:0 np:i:7 ql:Z:IIIIIIIIII qt:Z:IIIIIIIIII rq:f:0.999815
sn:B:f,17.2518,25.3332,6.55462,11.2707 we:i:8638323 ws:i:40959 zm:i:261297081 qs:i:26 qe:i:17132 bc:B:S,0,0 bq:i:100 cx:i:12
ls:B:C,134,162,98,99,164,48,45,45,48,162,98,113,100,164,108,101,97,100,130,162,113,108,186,73,73,73,73,73,73,73,73,73,73,73,73,73,
73,73,73,73,73,73,162,115,113,186,65,65,71,67,65,71,84,71,71,84,65,84,67,65,65,67,71,67,65,71,65,71,84,65,67,84,166,110,101,115,116,101,100
,192,164,112,53,109,99,0,165,116,114,97,105,108,130,162,113,108,186,73,
,162,115,113,186,65,71,84,65,67,84,67,84,71,67,71,84,84,71,65,84,65,67,67,65,67,84,71,67,84,84 RG:Z:f56a67e5/0--0

PacBio mapping – Minimap2

Minimap2 <https://github.com/lh3/minimap2>

Uses **minimizers** – short sequences of length $-k$ [default 15 bases]

1. Extracts minimizers from the reference (target) and index them
2. Match each minimizer in the query sequence against the reference set of minimizers
3. Sorts the minimizers after genomic position
4. Make a chain of overlapping minimizers
5. Repeat for all query sequences

```
minimap2 -ax map-hifi ref.fa query.bam > output.sam # Pacbio HiFi
```

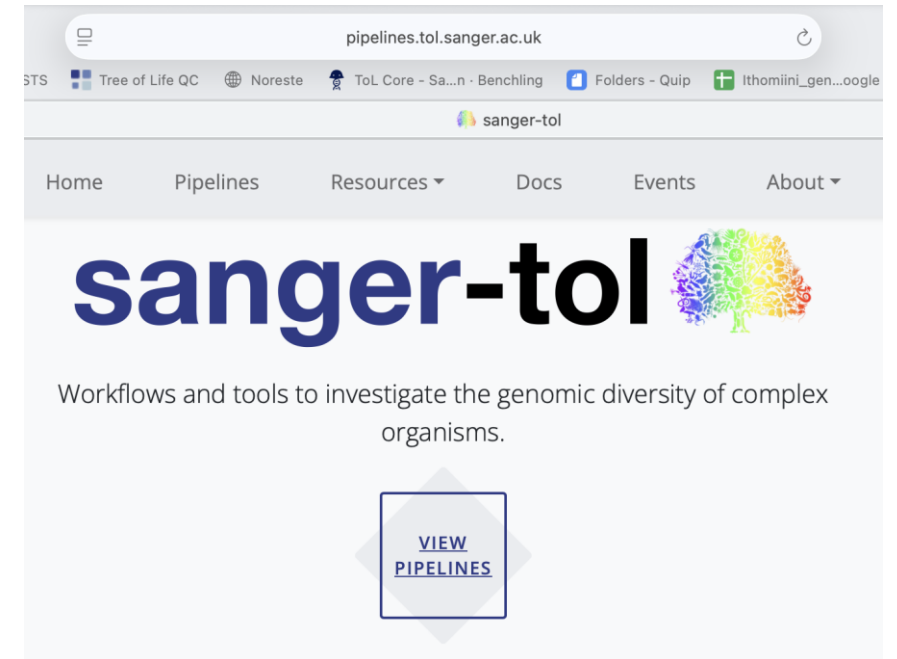
```
minimap2 -ax map-ont ref.fa ont.fq.gz > output.sam # Oxford Nanopore
```


Readmapping pipeline

Preprocessing - filtering
and adaptor removal

Alignment - minimap2

Alignment postprocessing
- mapping statistics



<https://pipelines.tol.sanger.ac.uk>

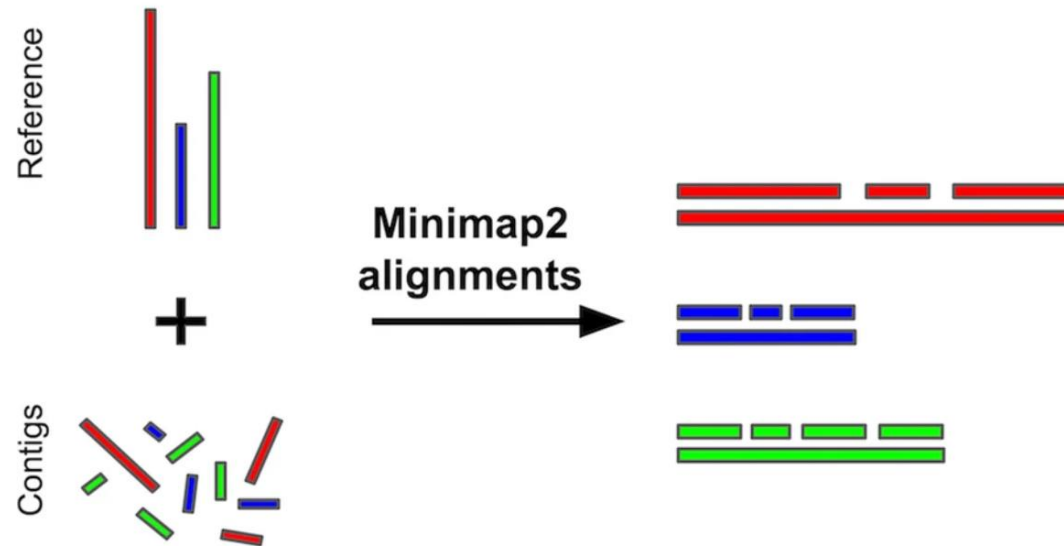
Generating and finding
reference genomes

Improving a draft genome assembly with a chromosome-level reference genome of a close relative

Reference-guided scaffolding with RagTag (successor of RAGOO)

<https://github.com/malonge/RagTag>

Uses minimap2 for genome alignment



Generating reference genomes with long reads and Hi-C data

Contigs from assembled long reads

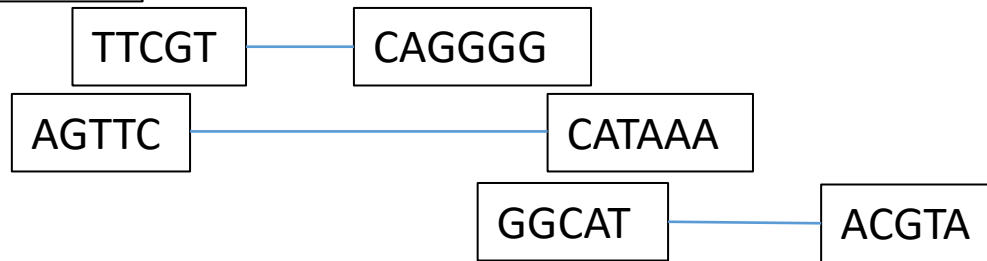
Contig 1

ATAGTGTGGATGCTGAGTTCGT



Contig 2

TTCAGGGGGCATAAAAGTAACGTA



Contig 3

GACACCCGCATATTAGTAAC



Contig 4

ATCGTA...



Hi-C data

Scaffold 2

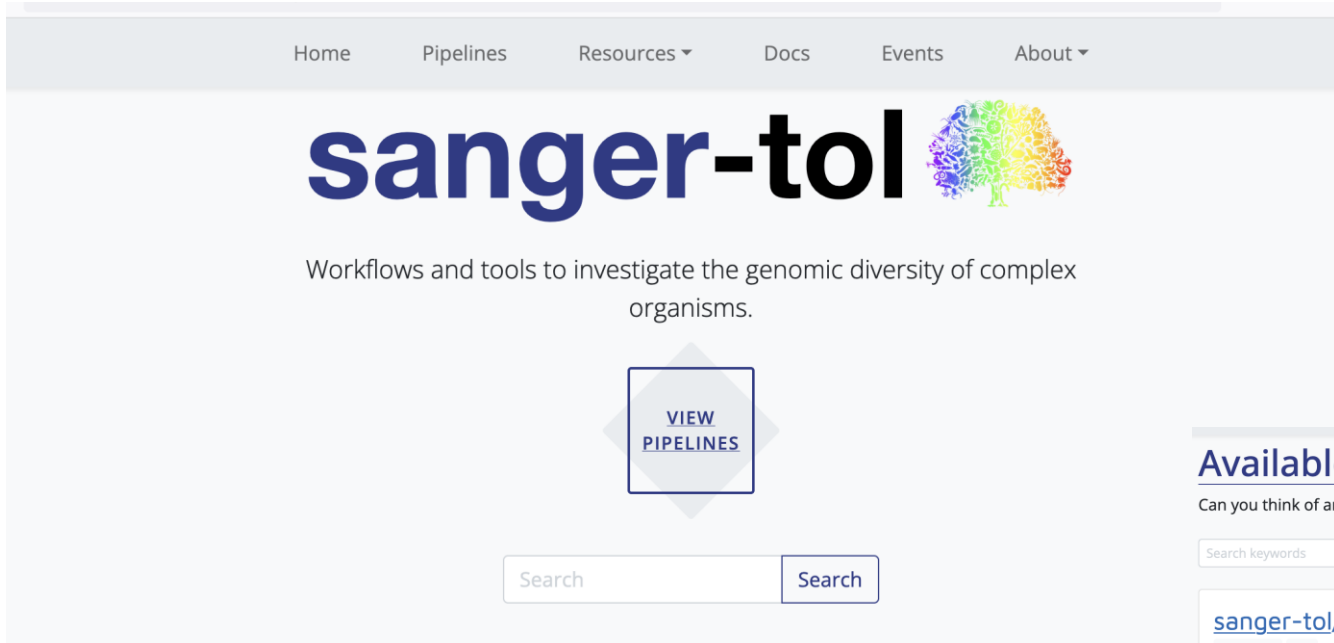
ATAGTGTGGATGCTGAGTTCGT **NNN** TTCAGGGGGCATAAAAGTAACGTA

Scaffold 2

GACACCCGCATATTAGTAAC **NNN** ATCGTA...

Pipelines to generate a genome

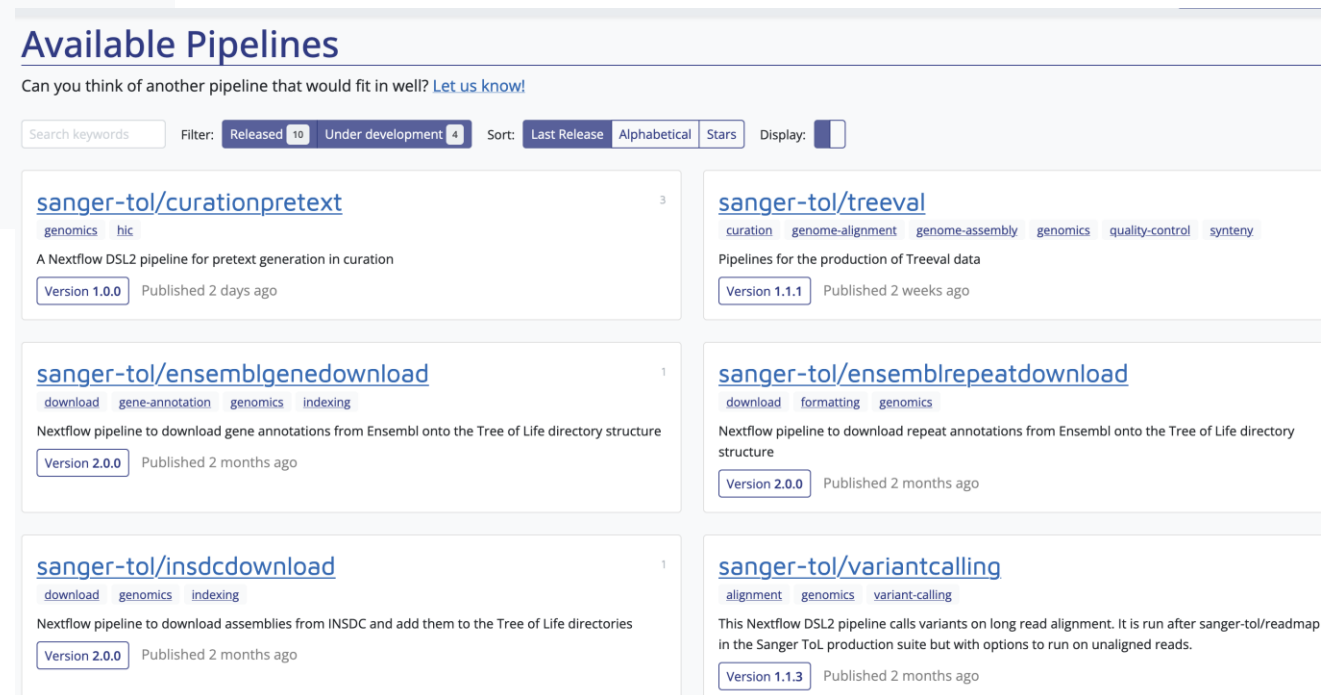
<https://pipelines.tol.sanger.ac.uk/>



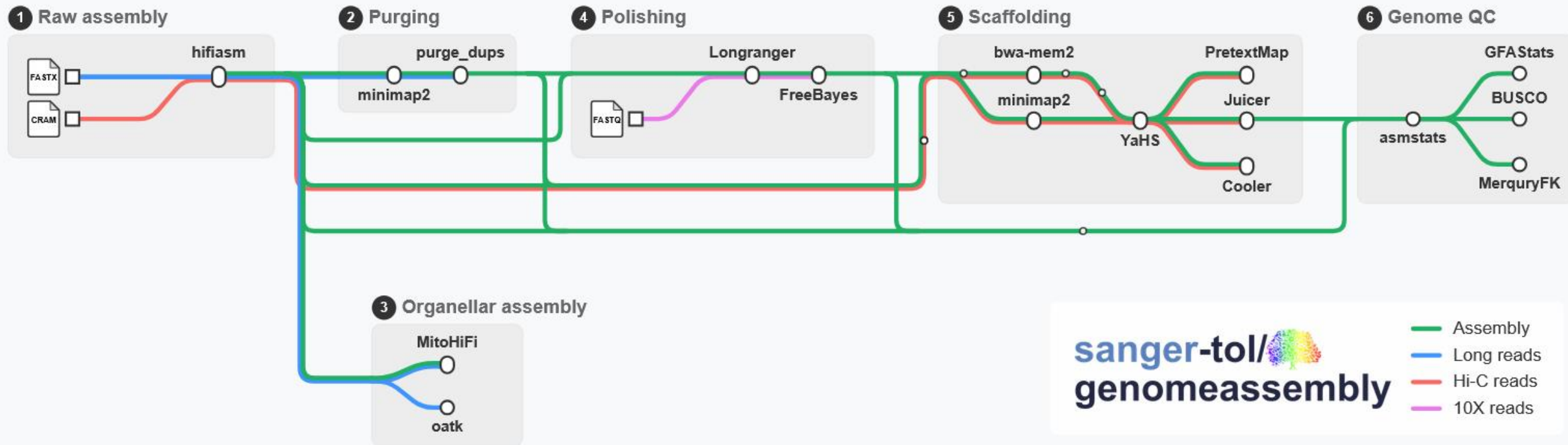
Sanger-Tol recipe is working
across the Tree of life
>4000 species already with
reference genomes

Chromosome level genomes

- 25x Pacbio HiFi
- 100x Hi-C (Arima/Qiagen)



Tree of Life pipeline for genome assembly



Checking if a species or genus has a genome

<https://goat.genomehubs.org>



Genomes on a Tree (GoaT)

Genomes on a Tree (GoaT): A versatile, scalable search engine for genomic and sequencing project metadata across the eukaryotic tree of life.

Challis *et al.* 2023. Wellcome Open Res 2023, 8 [doi.12688/wellcomeopenres.18658.1](https://doi.org/10.12688/wellcomeopenres.18658.1)

GoaT has been built using [GenomeHubs](#) to help coordinate efforts across the [Earth Biogenome Project](#) (EBP) Network at all stages from planning through sequencing and assembly to publication. [read more...](#)

Search GoaT

Type to search GoaT taxon index (e.g. Canidae)

Q TAXON



include descendants

☐ Off

include
estimates

☒ On

empty columns

☐ Off

result columns



query builder



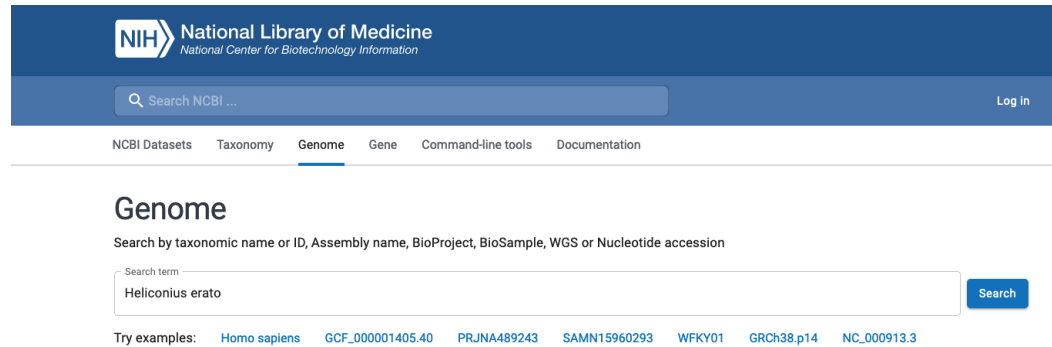
clear all



Where can we find genomes?

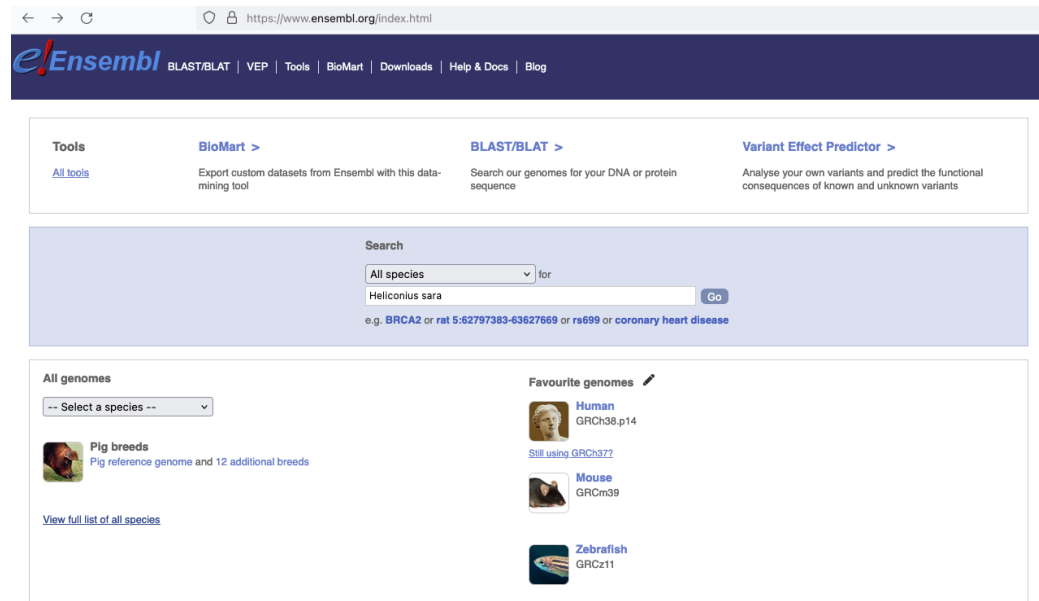
- NCBI (National Center for Biotechnology Information):

<https://www.ncbi.nlm.nih.gov/datasets/genome/>



- ensembl

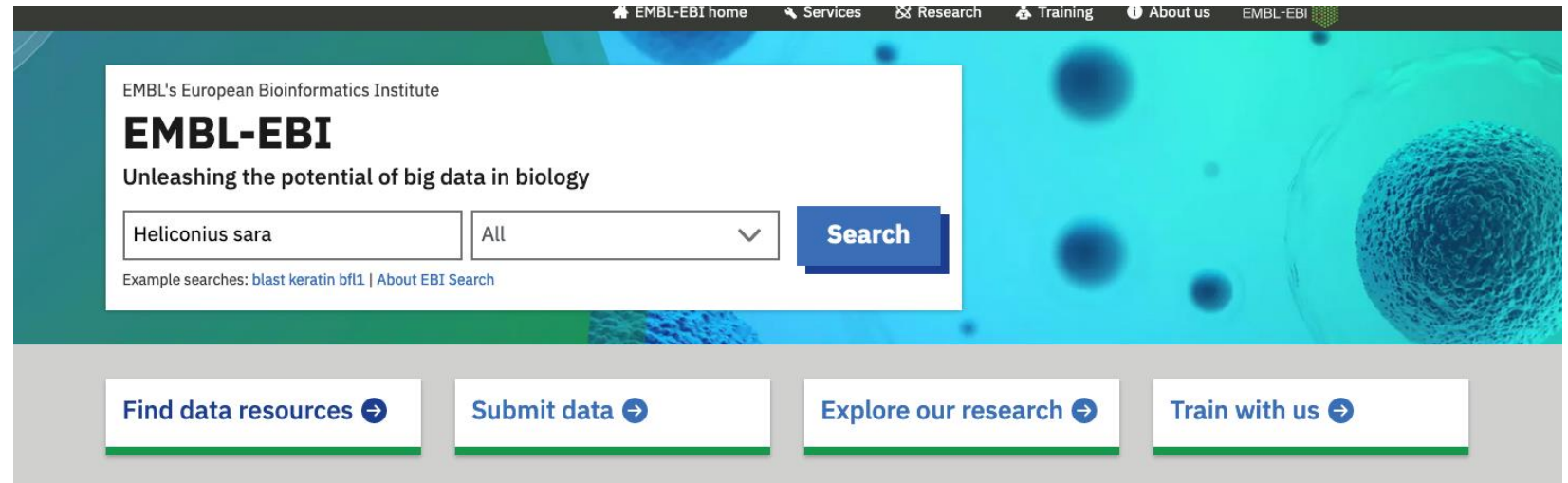
<https://www.ensembl.org/index.html>



Where can we find genomes?

- EMBL-EBI (European Molecular Biology Laboratory - European Bioinformatics Institute):

<https://www.ebi.ac.uk/>



Example

https://www.ncbi.nlm.nih.gov/datasets/genome/

An official website of the United States government [Here's how you know](#)

NIH

National Library of Medicine

National Center for Biotechnology Information

Search NCBI ...

Log in

NCBI DatasetsTaxonomyGenomeGeneCommand-line toolsDocumentation

Genome

Search by taxonomic name or ID, Assembly name, BioProject, BioSample, WGS or Nucleotide accession

Search term

Heliconius sara

×

Search

Try examples: [Homo sapiens](#) [GCF_000001405.40](#) [PRJNA489243](#) [SAMN15960293](#) [WFKY01](#) [GRCh38.p14](#) [NC_000913.3](#)

https://www.ncbi.nlm.nih.gov/datasets/genome/GCA_917862395.2/

An official website of the United States government [Here's how you know](#)

NIH

National Library of Medicine

National Center for Biotechnology Information

Search NCBI ...

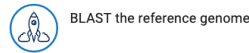
Log in

NCBI DatasetsTaxonomyGenomeGeneCommand-line toolsDocumentation

Genome assembly iHelSar1.2 reference

	Download	datasets	URL	FTP	Actions
Submitted GenBank assembly			GCA_917862395.2		⋮
Taxon			Heliconius sara		
WGS project			CAKJTV02		
Assembly type			haploid		
Submitter			WELLCOME SANGER INSTITUTE		
Date			Apr 6, 2023		

View the [legacy Assembly page](#)



Additional genomes

[Browse all Heliconius sara genomes \(2\)](#)

BioProject

[PRJEB46449](#)
Heliconius sara (Sara longwing) genome assembly iHelSar1

https://www.ncbi.nlm.nih.gov/datasets/genome/?taxon=33443

An official website of the United States government [Here's how you know](#)

NIH

National Library of Medicine

National Center for Biotechnology Information

Search NCBI ...

Log in

NCBI DatasetsTaxonomyGenomeGeneCommand-line toolsDocumentation

Genome

Download a genome data package including genome, transcript and protein sequence, annotation and a data report

Selected taxa

Heliconius sara Enter one or more taxonomic names

Filters

Download Select columns

2 Genomes 1 selected

Rows per page 20 1-2 of 2

☐ Assembly	GenBank	RefSeq	Scientific name	Modifier	Annotation	Action
☒ iHelSar1.2	GCA_917862395.2		Heliconius sara			⋮
☐ iHelSar1.1 alternate haplotype...	GCA_911392485.1		Heliconius sara			⋮

https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/917/862/395/GCA_917862395.2_iHelSar1.2/

Index of /genomes/all/GCA/917/862/395/GCA_917862395.2_iHelSar1.2

Name	Last modified	Size
Parent Directory		-
GCA_917862395.2_iHelSar1.2_assembly_report.txt	2023-04-18 17:05	34K
GCA_917862395.2_iHelSar1.2_assembly_stats.txt	2023-04-18 17:05	31K
GCA_917862395.2_iHelSar1.2_feature_count.txt	2024-02-15 01:58	168
GCA_917862395.2_iHelSar1.2_genomic.fna.gz	2023-04-18 17:05	107M
GCA_917862395.2_iHelSar1.2_genomic.gbff.gz	2023-04-18 17:05	138M
GCA_917862395.2_iHelSar1.2_genomic_gaps.txt.gz	2023-04-18 17:05	1.4K
GCA_917862395.2_iHelSar1.2_wgsmaster.gbff.gz	2023-04-18 17:05	1.4K
README.txt	2024-04-11 16:11	54K
annotation_hashes.txt	2024-02-17 10:02	410
assembly_status.txt	2024-07-15 13:53	14
md5checksums.txt	2024-02-17 10:02	625
uncompressed_checksums.txt	2024-06-14 01:21	167

[HHS Vulnerability Disclosure](#)

wget
https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/917/862/395/GCA_917862395.2_iHelSar1.2/GCA_917862395.2_iHelSar1.2_genomic.fna.gz

Reference genomes are stored in a fasta format

It is a text-based format for representing nucleotide sequences (DNA or RNA)

```
>NG_008679.1:5001-38170 Homo sapiens paired box 6 (PAX6) ← Description Line  
ACCTCTTTTCTTATCATTGACATTTAAACTCTGGGGCAGGTCCTCGCGTAGAACGCGGCTGTCAGATCT  
GCCACTTCCCCTGCCGAGCGGCGGTGAGAAGTGTGGGAACCGGCGCTGCCAGGCTCACCTGCCTCCCCGC  
CCTCCGCTCCCAGGTAACCGCCCCGGGCTCCGGCCCCGGCCCCGGCTCGGGGCCCCGCGGGGCCTCTCCGCTG  
CCAGCGACTGCTGTCCCCAAATCAAAGCCCCGCCCAAGTGGCCCCGGGGCTTGATTTTTTGCTTTTAAAAG  
GAGGCATACAAAGATGGAAGCGAGTTACTGAGGGAGGGATAGGAAGGGGGGTGGAGGAGGGACTTGTCTT  
TGCCGAGTGTGCTCTTCTGCAAAAGTAGCAAAATGTTCCACTCCTAAGAGTGGACTTCCAGTCCGGCCCT ← Sequence Lines  
GAGCTGGGAGTAGGGGGCGGGAGTCTGCTGCTGTCTGCTAAAGCCACTCGCGACCGCGAAAAATGCA  
GGAGGTGGGGACGCACTTTGCATCCAGACCTCCTCTGCATCGCAGTTCACGACATCCACGCTTGGGAAAG  
TCCGTACCCGCGCCTGGAGCGCTTAAAGACACCCTGCCGCGGGTCGGGCGAGGTGCAGCAGAAGTTTCCC  
GCGGTTGCAAAGTGCAGATGGCTGGACCGCAACAAAGTCTAGAGATGGGGTTCGTTTCTCAGAAAGACGC
```

Description Line: Begins with a > character followed by an identifier and optional description.

Sequence Lines: The actual sequence data, typically represented over multiple lines for readability.

Assembly statistics

	GenBank
Genome size	562.2 Mb
Total ungapped length	562.2 Mb
Number of contigs	203,084
Contig N50	9.9 kb
Contig L50	14,088
GC percent	33.5
Genome coverage	1.9x
Assembly level	Contig

It is the median of the contig lengths.

It is the number of contigs (or scaffolds) that together contain at least 50% of the total assembly length.

Significance

N50: A higher N50 value indicates that the assembly contains longer contigs, suggesting better assembly quality and continuity.

L50: A lower L50 value indicates that fewer contigs are needed to cover 50% of the genome, suggesting higher assembly contiguity.

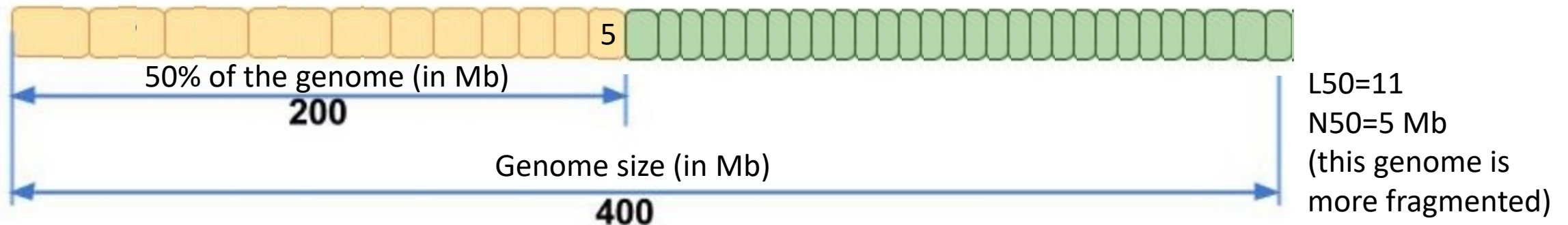
L50: number of largest contigs/scaffolds arranged by size that make up 50% of the genome. Ideally, it is close to half the number of chromosomes.

N50: length of the contig/scaffold where 50% of the genome is contained in contigs/scaffolds longer than this contig/scaffold, and 50% shorter than this contig, more or less the median contig length. Ideally it is close to the median chromosome length.

Nice genome (ideally the number of scaffolds should be the number of chromosomes)



More fragmented genome



Note, the ideal values of L50 and N50 depend on the genome size and number of chromosomes. The «more fragmented genome» could also just stem from a species with more chromosomes.

Now, let's align our filtered
and trimmed reads to the
reference genome

Aligning the reads with bwa-mem2

1. Index the reference genome (to speed up the alignment)
2. Align reads to the reference with bwa-mem2 to generate a sam file with read groups (read groups specify the individual, sequencer, etc)
3. Convert the sam file to a compressed bam file with samtools
4. Sort the bam file with samtools
5. Index the bam file with samtools
6. Remove duplicates (e.g. PCR duplicates or technical artefacts) with picardtools
7. Index the new bam file with samtools

Bam files

= binary (compressed) version of sam files that are much smaller than sam files without losing information (note, cram files are even more efficiently compressed. If you have large datasets and want to keep these files, consider converting the sam/bam to cram files, which are more space-efficient)

Header lines start with @

```
@HD      VN:1.0  S0:coordinate
@SQ      SN:chr20      LN:64444167
@PG      ID:TopHat      VN:2.0.14      CL:/srv/dna_tools/tophat/tophat -N 3 --read-edit-dist 5 --read-rea
lign-edit-dist 2 -i 50 -I 5000 --max-coverage-intron 5000 -M -o out /data/user446/mapping_tophat/index/chr
20 /data/user446/mapping_tophat/L6_18_GTGAAA_L007_R1_001.fastq
HWI-ST1145:74:C101DACXX:7:1102:4284:73714      16      chr20      190930      3      100M      *      0      0
      CCGTGTTTAAAGGTGGATGCGGTCACCTTCCCAGCTAGGCTTAGGGATTCTTAGTTGGCCTAGGAAATCCAGCTAGTCCTGTCTCTCAGTCCCCCCTCT
C      BBDCCDDCCDDDDDCDDDDDDDCDDCCDBC?DDDDDDDDDDDDDDDDCCDDDDDDDDDDDDCCCCEDDDC?DDDDDDDDDDDDDDDDDDDDDBDHFFFFDC@@
      AS:i:-15      XM:i:3  XO:i:0  XG:i:0  MD:Z:55C20C13A9 NM:i:3  NH:i:2  CC:Z:=  CP:i:55352714  HI:i:0
HWI-ST1145:74:C101DACXX:7:1114:2759:41961      16      chr20      193953      50      100M      *      0      0
      TGCTGGATCATCTGGTTAGTGGCTTCTGACTCAGAGGACCTTCGTCCCCTGGGCGAGTGACCTTCCAGTGATTCCCTTGACATAAGGGGCATGGACGA
G      CDDDDDEDDDDDDDCDDDDDDDCDDDDCDDDDDEEC-DFFFEJJJJJIGJJJJJHGBHHGJIJJJJJJGJJJJJJJJJJHJJJJJHHHHHFFFFFCCC
      AS:i:-16      XM:i:3  XO:i:0  XG:i:0  MD:Z:60G16T18T3 NM:i:3  NH:i:1
```

Read
name

Sequence

SAM flag

(<https://broadinstitute.github.io/picard/explain-flags.html>)

Mapping start position

Mapping information (CIGAR string)

Removing duplicates with picard tools

- For whole-genome resequencing with paired-end Illumina data, we can detect PCR duplicates or technical duplicates by removing read pairs that start and end at exactly the same positions as another read pair. As for whole-genome resequencing, the DNA fragments start and end at random positions, any read pairs that start and end at the exact same positions are likely duplicates of the same DNA fragments and should be removed.
- Note: if you have ddRAD, GBS data, or amplicons where it is normal that they start and end at the same position, DO NOT REMOVE DUPLICATES.

Reference genome



Likely PCR duplicates as they start and end at the same position
-> Keep only one



ddRAD data of the same locus
all start and end at the same position
-> Do not remove «duplicates»